CAIRO UNIVERSITY · FACULTY OF NURSING · CRITICAL CARE & EMERGENCY NURSING DEPT

Assessment of *Critically Ill Patients*

Comprehensive Reference on Critical Care Definitions, ABCDE Primary Survey, Four Assessment Stages (Pre-Arrival, Admission Quick Check, Comprehensive Admission, Ongoing), Neurological (AVPU, GCS, PERRLA, BPS, CPOT, RASS, CAM-ICU), Cardiovascular (5 Ps, DVT, JVP "POLICE", PQRST), Respiratory Patterns, Renal, GI, and Integumentary (Braden, PUSH, SKIN Bundle) Assessments.

Booklet Summary · **Critical Care & Emergency Nursing Department** · *Cairo University*

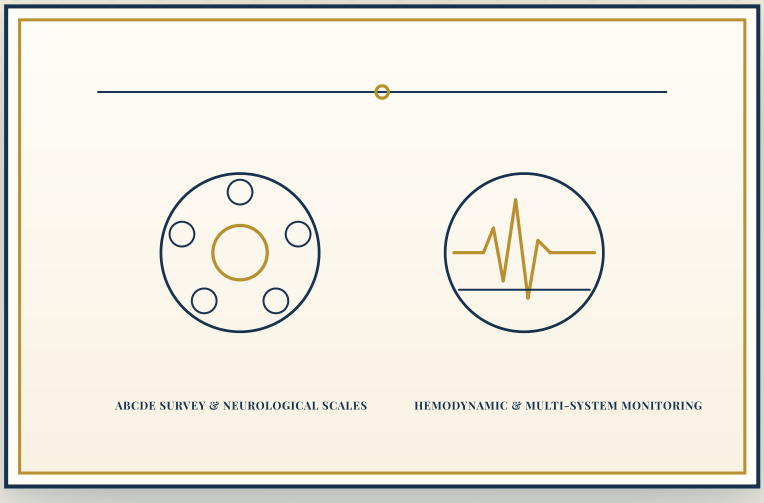


PLATE I — *Rapid ABCDE primary survey and neurological evaluation (left), paired with continuous ICU hemodynamic, respiratory, and organ-perfusion monitoring (right).*

Duas Before Studying

ابدأ مذاكرتك بهذه الأدعية المباركة لتيسير الفهم والحفظ والعلم النافع:

دعاء الفهم

اللَّهُمَّ إِنِّي أَسْأَلُكَ فَهْمَ النَّبِيِّينَ، وَحِفْظَ الْمُرْسَلِينَ، وَالْمَلَائِكَةِ الْمُقَرَّبِينَ

دعاء قبل المذاكرة

اللَّهُمَّ افْتَحْ عَلَيَّ فُتُوحَ الْعَارِفِينَ بِحِكْمَتِكَ، وَيَسِّرْ لِي أَمْرِي، وَاشْرَحْ لِي صَدْرِي بِمَعْرِفَتِكَ

دعاء العلم النافع

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا، وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا

دعاء سيدنا موسى عليه السلام

رَبِّ اشْرَحْ لِي صَدْرِي، وَيَسِّرْ لِي أَمْرِي، وَاحْلُلْ عُقْدَةً مِنْ لِسَانِي يَفْقَهُوا قَوْلِي

نية المذاكرة

نَوَيْتُ بِمُذَاكَرَتِي هَذِهِ خِدْمَةَ مَرْضَى الْمُسْلِمِينَ وَنَفْعَ أُمَّةِ نَبِيِّكَ مُحَمَّدٍ ﷺ

دعاء تيسير العلم

اللَّهُمَّ لَا سَهْلَ إِلَّا مَا جَعَلْتَهُ سَهْلًا، وَأَنْتَ تَجْعَلُ الْحَزْنَ إِذَا شِئْتَ سَهْلًا

إرشاد للمذاكرة: يغطي هذا الملخص كتاب قسم تمريض الحالات الحرجة والطوارئ بجامعة القاهرة (**تقييم المرضى البالغين ذوي الحالات الحرجة والحادة — Assessment of Critically Ill Patients**) بالكامل طبقًا للنص الرسمي حرفيًا مع جميع الجداول والمقاييس الإكلينيكية.

1. Key Definitions, Assessment Approaches & Stages 1–2 (ABCDE)

1. Introduction, Key Definitions & Assessment Approaches

Introduction: Critical care assessment is vital because critically ill patients can **deteriorate rapidly**. A **systematic, organized approach** enables timely recognition of life-threatening changes and guides lifesaving interventions. Assessment of critically ill patients is an **essential competency** for critical care nurses, and effective assessment forms the **foundation for patient management** in critical care settings.

KEY DEFINITIONS

— **Critical care nursing:** Specialized nursing practice focused on caring for patients with life-threatening conditions; involves working with physicians and specialists to assess, treat, and monitor critically ill patients while providing their basic care.

— **Critically ill patient:** A patient who has a life-threatening condition requiring intensive monitoring and advanced organ support (e.g., **mechanical ventilation, vasoactive medications**) without which death would be imminent.

— **Critical care environment:** A high-technology intensive care setting (e.g., **ICU**) with specialized staff and equipment, where patients receive continuous monitoring and care; a dynamic system in which information flows constantly.

— **Assessment (nursing):** Systematic and continuous collection of patient data (**subjective and objective**) covering physiological, psychological, social, and spiritual needs; the **first step in the nursing process**.

ASSESSMENT APPROACHES & 4 ASSESSMENT STAGES

— **Traditional/systematic assessment:** Head-to-toe or body-systems examination examining each organ system sequentially (*History* → *Examination* → *Differential* → *Investigations* → *Diagnosis* → *Treatment*).

— **ABCDE approach:** Used in emergencies and initial encounters for a rapid primary survey: **A**irway (& oxygenation), **B**reathing (& ventilation), **C**irculation (& shock management), **D**isability (neurological deterioration), and **E**xposure (& examination).

— **Four Assessment Stages:**
1. **Pre-Arrival Assessment**
2. **Admission Quick Check Assessment (Primary Survey)**
3. **Comprehensive Admission Assessment**
4. **Ongoing Assessment**

2. First: Pre-Arrival Assessment & Second: Admission Quick Check (Primary Survey)

FIRST: PRE-ARRIVAL & STANDARD ROOM SETUP (12 ITEMS)

— Begins when info is received about an impending ICU admission; nurse gathers details (**diagnosis, vitals, history**) and prepares equipment/staff:

— **1.** Bedside ECG and invasive pressure monitor with cables | **2.** ECG electrodes | **3.** BP cuff | **4.** Pulse oximetry | **5.** Suction gauges & canister setup | **6.** Suction catheters | **7.** Bag-valve mask device | **8.** O_2 flow meter, tubing & delivery device | **9.** IV poles & infusion pumps | **10.** Bedside supply cart (alcohol swabs, nonsterile gloves, syringes, chux, dressing supplies) | **11.** Admission kit (bath basin & hygiene supplies) | **12.** Admission & critical care documentation forms.

SECOND: ADMISSION QUICK CHECK (ABCDE SUMMARY)

— **Airway:** Patency, position of artificial airway (if present).

— **Breathing:** Effectiveness, rate, depth, pattern, symmetry, effort, accessory muscles, breath sounds, chest tube fluctuation, synchrony with MV & spontaneous breathing.

— **Circulatory status:** ECG (rate, rhythm, ectopy), BP, peripheral pulses, capillary refill, skin (color, temp), active bleeding & LOC (**AVPU**).

— **Disability, Drugs & Diagnostic tests:** Start I&O, prior drugs (prescribed, OTC, illicit) & review labs. **Common ICU investigations: Serum electrolytes, Coagulation studies, CBC, ABG, Glucose, Chest x-ray, ECG**.

— **Exposure & Equipment:** Full body for wounds/rashes; check vascular/drainage tubes for location, patency, functioning & labeling.

INITIAL ASSESSMENT (ABCDE)	MEASURE	ACTION	CONSIDER (AFTER INITIAL)
A – Airway: Is airway patent – can patient talk? Snoring, stridor, obstruction (foreign body, vomit, blood, edema), cervical spine	—	**Non-patent airway:** Head tilt, chin lift, jaw thrust; Suction; Naso/oropharyngeal airway; **O_2 (15 L/min)**	—
B – Breathing: Cyanosis, use of accessory muscles, breathing depth & rhythm, tracheal position, symmetrical chest expansion, breath sounds & auscultation, chest percussion	• Respiratory rate • SpO_2	• Positioning of patient • Bag/pocket mask ventilation • Decompression of pneumothorax • Inhalations	• ABG • Chest X-ray
C – Circulation: Bleeding; Skin: color (pale, red, mottled), cool/warm/dry/sweaty; Auscultation	• Capillary refill time • Pulse & BP • ECG	• Stop bleeding • IV/IO access • Fluids/blood	• 12-lead ECG & Blood tests • Urinary catheter • ECHO/FAST/FATE
D – Disability: AVPU; Pupils (reaction, size, equal); Neck stiffness	• GCS • Blood glucose	• Recovery position	• Lumbar puncture • Focused neurologic assessment • Rectal exam (sphincter tonus)
E – Exposure: Head-to-toe: Trauma, fractures, wounds, lesions, bleeding, infection, petechiae, rash	• Temperature	• Prevent hypo-/hyperthermia • Stabilize fracture	• Blood cultures • Culture from wound • Antibiotics

التعريفات، مداخل التقييم، والمرحلتان الأولى والثانية (Pre-Arrival & Admission Quick Check ABCDE):

• **التعريفات ومداخل التقييم:** المريض ذو الحالة الحرجة يعاني من حالة تهدد الحياة تتطلب مراقبة مكثفة ودعمًا متقدمًا للأعضاء (مثل التنفس الصناعي والأدوية القابضة للأوعية). ويتم التقييم إما بالطريقة التقليدية المنظمة (من الرأس للقدمين) أو بمدخل **ABCDE** السريع في الطوارئ.

• **مراحل التقييم الأربع:** (1) **تقييم ما قبل الوصول (Pre-Arrival):** جمع معلومات التشخيص والعلامات الحيوية وتجهيز الغرفة بـ 12 أداة قياسية؛ (2) **الفحص السريع عند الدخول (Admission Quick Check):** تقييم **ABCDE** الفوري، ومراجعة الفحوصات الشائعة في العناية المركزة (الإلكتروليتات، سيولة الدم، صورة الدم الكاملة CBC، غازات الدم الشرياني ABG، السكر، أشعة الصدر، ورسم القلب ECG)؛ (3) **التقييم الشامل عند الدخول**؛ و(4) **التقييم المستمر (Ongoing Assessment).**

2. Comprehensive Admission Assessment & Neurological System (Part I)

1. Third: Comprehensive Admission Assessment (History, Psychosocial, Spiritual & SAMPLE)

Performed **after stabilization**: includes a detailed review of medical/social history, psychological/spiritual state, lab/imaging results, and a full head-to-toe physical examination to formulate nursing diagnoses (e.g., *ineffective airway clearance, impaired gas exchange, decreased cardiac output, acute confusion*).

1. PAST MEDICAL HISTORY (SAMPLE) & 2. SOCIAL HISTORY

— **Past Medical History:** Medical conditions & surgical procedures; previous hospitalizations; current medications (name, dose, route, frequency, time of last dose); last oral intake; history of falls; allergies (medication, food, latex, type of reaction).

— **SAMPLE Mnemonic: S** = Symptoms | **A** = Allergies | **M** = Medications | **P** = Pertinent Medical History | **L** = Last Oral Intake | **E** = Events leading to the illness.

— **Social History:** Age & gender, height & weight, education level, occupation/employment status, marital status, primary family members/significant others, religious affiliation, substance use (tobacco, alcohol, other).

3. PSYCHOLOGICAL, 4. SPIRITUALITY & 5. PHYSICAL OVERVIEW

— **Psychological Assessment:** General communication, psychiatric/emotional problems, coping styles, anxiety and stress, expectations of the ICU, current stresses (financial, family, occupational, health-related), and family needs (info, communication, emotional support, visitation).

— **Spirituality:** Faith/spiritual preference; healing practices (spiritual, cultural, or complementary practices).

— **Physical Assessment (7 Systems):** Nervous, Cardiovascular, Respiratory, Renal, Gastrointestinal, Endocrine/hematologic/immune, and Integumentary systems.

2. I. Neurological System: A. AVPU, B. Glasgow Coma Scale (GCS) & C. Pupil Assessment (PERRLA)

A. AVPU & B. GLASGOW COMA SCALE (GCS = E + V + M)

— **AVPU: A – Alert** (awake and alert) | **V – Verbal stimuli** (responds to verbal stimulation) | **P – Painful stimuli** (responds only to painful stimulation) | **U – Unresponsive** (does not respond to verbal or painful stimulation).

— **GCS Components (Total Score = 3 to 15):**
· **Eye Opening (E, 1–4):** Spontaneous (**4**), To verbal command (**3**), To pain (**2**), No response (**1**).
· **Verbal Response (V, 1–5):** Oriented (**5**), Confused (**4**), Inappropriate words (**3**), Incomprehensible sounds (**2**), No response (**1**).
· **Motor Response (M, 1–6):** Obeys commands (**6**), Localizes pain (**5**), Withdrawal from pain (**4**), Abnormal flexion / Decorticate (**3**), Extension / Decerebrate (**2**), No response (**1**).

C. PUPIL REACTION ASSESSMENT (PERRLA)

— **Normal pupil size:** Ranges from **2 to 4 mm** in diameter.

— **PERRLA Documentation Means:**
· **P – Pupils:** Both eyes are being examined.
· **E – Equal:** Both pupils are equal in size (**2 – 4 mm**).
· **R – Round:** Both pupils have a round, regular shape.
· **R – Reactive to L – Light:** Pupils constrict briskly when light is shone (direct and consensual response).
· **A – Accommodation:** Pupils constrict when focusing on a near object and dilate when focusing on a distant object.

ABNORMAL PUPIL FINDING	DESCRIPTION	POSSIBLE CAUSE / CLINICAL SIGNIFICANCE
Bilateral pinpoint pupils	Very small, equal pupils	**Opioid overdose, pontine lesion**
Bilateral fixed and dilated	Large, non-reactive pupils	**Severe brain injury, brain death, cardiac arrest**
Unilateral fixed and dilated	One large non-reactive pupil	**Cranial nerve III compression, transtentorial herniation**
Sluggish pupils	Slow reaction to light	**Raised ICP, metabolic disorders**
Hippus Phenomenon	Rhythmic dilation and constriction	**Early raised ICP, metabolic causes**
Oval pupils	Oval shaped pupil	**Early transtentorial herniation**

التقييم الشامل عند الدخول وتقييم الجهاز العصبي (AVPU, GCS, PERRLA):
• **التقييم الشامل (Comprehensive Admission Assessment):** يُجرى **بعد استقرار حالة المريض**، ويشمل التاريخ الطبي باختصار **SAMPLE**، والتاريخ الاجتماعي والنفسي والروحي، والفحص البدني الشامل.
• **مقياس جلاسكو للغيبوبة (GCS = E4 + V5 + M6) وتقييم الحدقتين (PERRLA):** يُحسب مجموع GCS من فتح العين (1–4) والاستجابة اللفظية (1–5) والحركية (1–6: لاحظ أن **3** = **انثناء غير طبيعي Decorticate** و**2** = **بسط غير طبيعي Decerebrate**). الحجم الطبيعي للحدقة **2–4 مم**، وتوثق الحالة الطبيعية باختصار **PERRLA**. ومن أهم العلامات غير الطبيعية: الحدقات الدبوسية الثنائية (**تسمم الأفيونات Opioid overdose أو إصابة الجسر Pontine lesion**)، واتساع وثبات حدقة واحدة (**انضغاط العصب الثالث CN III أو الفتق عبر الخيمة Transtentorial herniation**).

3. Neurological System (Part II): Pain, RASS, CAM-ICU, Motor & Reflexes

1. D. Pain Assessment in Critical Care (NRS, BPS & CPOT)

— **1. Numerical Rating Scale (NRS):** Patient rates pain on a scale from **0 to 10** (**0** = no pain, **1–3** = mild, **4–6** = moderate, **7–10** = severe / **10** = worst possible pain).

2. BEHAVIORAL PAIN SCALE (BPS)

— Assesses pain through observable behaviors across **3 domains**, each scored **1 to 4** (Total range **3–12**; **BPS score ≥ 6 is significant**):

— **1) Facial expression: 1** Relaxed | **2–3** Partially/fully tightened, grimacing | **4** Grimacing.

— **2) Upper limbs: 1** No movement | **2–3** Partially/fully bent with finger flexion / permanently retracted.

— **3) Compliance with ventilation: 1** Tolerating movement | **2–3** Coughing but tolerating / fighting ventilator | **4** Unable to control ventilation.

3. CRITICAL CARE PAIN OBSERVATION TOOL (CPOT)

— Evaluates **4 behavioral indicators** (Scored **0** vs. **1–2**):

— **1) Facial expression: 0** Relaxed/neutral | **1–2** Tense or grimacing.

— **2) Body movements: 0** Absence of movements | **1–2** Protection or restlessness.

— **3) Muscle tension: 0** Relaxed; no resistance to passive movement | **1–2** Tense/rigid or very tense/rigid.

— **4) Ventilator compliance (intubated) / Vocalization (extubated): 0** Tolerating ventilator or talking in normal tone | **1–2** Coughing/fighting ventilator or sighing/moaning/crying out.

2. E. Richmond Agitation and Sedation Scale (RASS) & F. Delirium (CAM-ICU)

SCORE	RASS LEVEL	EXACT CLINICAL DESCRIPTION
+4	**Combative**	Violent; immediate danger to staff
+3	**Very agitated**	Pulls at or removes tubes; aggressive
+2	**Agitated**	Frequent non-purposeful movements; fights ventilator
+1	**Restless**	Anxious/apprehensive, but movements are not aggressive or vigorous
0	**Alert and calm**	Alert and calm
−1	**Drowsy**	Not fully alert; sustained awakening to voice with eye opening and contact > **10 seconds**
−2	**Light sedation**	Briefly awakens to voice with eye opening and contact < **10 seconds**
−3	**Moderate sedation**	Movement or eye opening to voice, **without eye contact**
−4	**Deep sedation**	No response to voice, but movement or eye opening to **physical stimulation**
−5	**Unarousable**	No response to voice or physical stimulation

F. Delirium Assessment — CAM-ICU (4 Features Flowsheet):
Feature 1 (Acute Change or Fluctuating Course of Mental Status) + Feature 2 (Inattention: > 2 Errors on SAVEAHAART hand-squeeze or Pictures) PLUS either **Feature 3 (Altered LOC: Current RASS other than zero)** OR **Feature 4 (Disorganized Thinking: > 1 Error on 4 Yes/No questions + finger command when RASS = 0) → CAM-ICU Positive (Delirium Present)**.

3. G. Motor Function, Muscle Strength Grading (0 to +5) & Reflexes

MUSCLE STRENGTH GRADING SCALE (0 TO +5)

— Assess: **muscle size, muscle tone, muscle strength, movements, posture, and gait**:
· **Muscle size:** observe for symmetry or changes in muscle bulk.
· **Muscle tone:** resistance of muscles to movement.
· **Muscle strength:** ability to move against gravity and resistance.
· **Movements:** presence, symmetry, and quality of voluntary movements.
· **Posture:** body position and alignment.
· **Gait:** walking pattern and stability (when condition permits).

— **+5:** Full ROM, full strength (**Normal/full strength**).

— **+4:** Full ROM, less than normal strength (reduced strength).

— **+3:** Can raise extremity but **not against resistance** (movement against gravity only).

— **+2:** Can move extremity **but not lift it** (cannot overcome gravity).

— **+1:** Slight movement (minimal muscle activity).

— **0:** Complete paralysis (no observable movement).

DEEP TENDON & PLANTAR REFLEXES

— **Reflexes** are **involuntary motor responses to stimuli**; they provide information about the **integrity of the reflex arc** and nervous system.

— **Biceps Reflex (C5–C6):** Thumb over biceps tendon in antecubital fossa → **elbow flexion**.

— **Triceps Reflex (C7–C8):** Arm flexed 90°, tap tendon above olecranon → **elbow extension**.

— **Patellar Reflex (L2–L4):** Knee flexed ~30°, tap below patella → **knee extension**.

— **Plantar Response (Babinski Sign):** Normal = plantar flexion (toes flex downward); **Positive Babinski** = big toe dorsiflexes and other toes fan out (**abnormal in adults; indicates upper motor neuron lesion**).

— **Key points for ICU patients:** patient relaxed & limb supported; **compare both sides**; document response (**normal, decreased, absent, or exaggerated**); **absent reflexes** may occur with **deep sedation, neuromuscular blockers, peripheral neuropathy, or spinal cord injury**; sedation/analgesia/NMBs affect reflexes → assess **trends over time**.

تقييم الألم (NRS, BPS, CPOT)، التهدئة والهياج (RASS)، الهذيان (CAM-ICU)، والقوة العضلية:
• **مقاييس الألم:** مقياس **NRS (0–10)** للمريض القادر على التواصل؛ ومقياس **BPS (من 3 إلى 12، وتُعد الدرجة ≥ 6 ذات دلالة إكلينيكية)** عبر 3 محاور (تعبيرات الوجه، حركة الطرفين العلويين، والتوافق مع جهاز التنفس)؛ وأداة **CPOT** عبر 4 مؤشرات.
• **مقياس RASS (+4 إلى −5) وCAM-ICU:** يقيس RASS مستوى الهياج أو التهدئة (الهدف الطبيعي **Alert and calm = 0**؛ لاحظ الفرق بين **−1 تواصل بصري > 10 ثوانٍ** و**−2 تواصل بصري > 10 ثوانٍ** و**−3 حركة أو فتح عين للصوت دون تواصل بصري**). ويُشخَّص الهذيان بـ **CAM-ICU** بوجود تغير حاد/متذبذب (1) + عدم انتباه بأكثر من خطأين في (2) SAVEAHAART + إما RASS غير الصفر (3) أو تفكير غير منظم (4).
• **القوة العضلية والمنعكسات:** تُقيَّم القوة من **0 (شلل كامل)** إلى **+5 (قوة كاملة ضد الجاذبية والمقاومة)**، ووجود **علامة بابنسكي الإيجابية في البالغين** يدل على إصابة العصبون الحركي العلوي (**Upper motor neuron lesion**).

4. II. Cardiovascular System Assessment

1. Peripheral Perfusion, Pulses, Capillary Refill & Skin Inspection

A. PERIPHERAL PULSES (0 TO +4) & CAROTID PRECAUTIONS

— Check peripheral pulses and compare bilaterally.

— **Carotid Artery Precautions:** Avoid compression of the **carotid sinus**; **do not palpate both carotid arteries simultaneously**; auscultate for a **bruit** (blowing or swishing sound) and palpate for a **thrill** (vibrating sensation).

— **Pulse Strength Scale (0 to +4):**
· **0:** Absent pulse
· **+1:** Palpable but thready; easily obliterated with light pressure
· **+2:** Normal; cannot be obliterated with light pressure
· **+3:** Full | **+4:** Full and bounding

B. CAPILLARY REFILL & C. SKIN INSPECTION CLUES

— **Capillary Refill:** Apply pressure to nail bed until white; **failure of color to return within 3 seconds indicates impaired circulation**.

— **Skin Inspection Findings:**
· **Cool, clammy skin:** Vasoconstriction.
· **Warm, moist skin:** Vasodilation.
· **Flushing:** Medications, excess heat, anxiety, or fear.
· **Pallor:** Anemia or increased peripheral vascular resistance (atherosclerosis).
· **Dependent rubor (redness):** Chronic arterial insufficiency.
· **Peripheral cyanosis:** Bluish discoloration of lips and extremities.

2. D. Arterial Insufficiency (The 5 Ps), E. Venous Insufficiency (DVT) & F. JVD ("POLICE")

D. ARTERIAL ISCHEMIA — THE 5 PS

— **1. Pain:** Pain in affected limb.

— **2. Pulse:** Reduced or absent peripheral pulse.

— **3. Pallor:** Pale or abnormal color of limb.

— **4. Paresthesia:** Abnormal sensation (tingling or numbness).

— **5. Paralysis:** Loss or impairment of motor function.

E. VENOUS INSUFFICIENCY & DVT

— Veins (most commonly in legs) cannot pump blood back → blood pools; can lead to **fatal pulmonary embolism**.

— **ICU Risk Factors:** Immobility/bed rest, invasive lines/catheters, surgery/trauma, dehydration, underlying conditions.

— **Signs of DVT: Swelling, Warmth, Erythema (redness), Pain & Tenderness**.

F. ELEVATED JVP — "POLICE"

— **JVD** = visible bulging/distension of the jugular veins in the neck; occurs with **increased pressure in the venous system**; indicates the **heart is unable to handle the volume of blood returning to it**; reflects **elevated right atrial pressure** (assessed at **45°**); also called **Jugular Venous Pressure (JVP)**.

— **Causes of elevated JVP — "POLICE":**

— **P:** Pulmonary embolism / Pulmonary HTN

— **O:** Obstruction (SVC obstruction)

— **L:** Left and Right heart failure

— **I:** Increased fluid volume (overload)

— **C:** Cardiac tamponade / Constrictive pericarditis

— **E:** Emphysema / Cor pulmonale

3. G. Peripheral Pitting Edema Scale (0 to +4) & H. Chest Pain Assessment (PQRST)

G. GRADING PITTING EDEMA (0 TO +4)

— **0:** No depression in tissue.

— **+1 (2 mm):** Small depression in tissue, disappearing in < **1 second**.

— **+2 (4 mm):** Depression disappears in < **1–2 seconds**.

— **+3 (6 mm):** Depression disappears in < **2–3 seconds**.

— **+4 (8 mm):** Depression disappears in ≥ **4 seconds**.

H. CHEST PAIN ASSESSMENT — PQRST

— **P – Provoke:** What provokes/precipitates pain? (stairs, walking, or at rest).

— **Q – Quality:** Pressure or tightness; associated with nausea, vomiting, or diaphoresis.

— **R – Radiation:** Jaw, neck, scapular area, or left arm.

— **S – Severity:** Rate on a **1–10 scale** (10 = worst pain).

— **T – Timing:** Onset, duration, waxed/waned or constant, previous episodes.

تقييم الجهاز الدوري والقلب والأوعية الدموية (Cardiovascular Assessment):
• **النبض الطرفي والشريان السباتي وزمن امتلاء الشعيرات:** يُمنع فحص الشريانين السباتيين في نفس الوقت أو الضغط على الجيب السباتي (Carotid sinus)، ويُسمع الشريان السباتي لاكتشاف الـ **Bruit (صوت نفخة)** ويُحَس لاكتشاف الـ **Thrill (اهتزاز)**. وتُقاس قوة النبض من **0 إلى +4 (الطبيعي +2)**، وتأخر عودة لون فراش الظفر لأكثر من **3 ثوانٍ** يدل على ضعف التروية.
• **قصور الشرايين (Ps 5)، جلطة الأوردة العميقة (DVT)، واحتقان أوردة الرقبة (POLICE):** علامات نقص التروية الشريانية الخمس هي (**Pain, Pulse, Pallor, Paresthesia, Paralysis**)؛ وعلامات **DVT** هي التورم والدفء والاحمرار والألم والإيلام عند اللمس وقد تؤدي إلى **جلطة رئوية قاتلة**. ويعكس انتفاخ الوريد الوداجي **JVD/JVP** عند زاوية 45°) ارتفاع ضغط الأذين الأيمن وأسبابه تُجمع في كلمة **POLICE**.
• **تقييم الوذمة الانطباعية (Pitting Edema 0 to +4) وألم الصدر (PQRST):** تُصنف الوذمة حسب عمق الانطباع وزمن اختفائه (من **+1 أقل من ثانية** إلى **+4 أربع ثوانٍ أو أكثر**)، ويُقيّم ألم الصدر بمنهجية **PQRST**.

5. III. Respiratory System & IV. Renal System Assessment

1. III. Respiratory System: Airway, Breathing, Cyanosis/Clubbing & Abnormal Patterns

A–D. AIRWAY, BREATHING & CHEST ASSESSMENT

— **A. Artificial Airway:** Assess **tube placement** (position), **tube security**, and **cuff status**.

— **B. Airway Patency:** Evaluate **airway secretions** (suction when required) and adequacy of **humidification**.

— **C. Breathing Assessment:** Respiratory rate, volume, and pressure; **ABG analysis**; **Pulse oximetry**; **Capnograph**.

— **D. Chest Assessment:** Chest sounds, symmetry of chest wall, use of accessory muscles, skin color & nail beds, clubbing, abnormal patterns.

E. SKIN COLOR & NAIL BEDS (CYANOSIS & CLUBBING)

— **Cyanosis:** A **late sign of hypoxemia**. The **tongue and mucous membranes** are the most reliable sites for observing cyanosis.

— **Clubbing:** A possible sign of **long-term hypoxemia**; observe the shape of the nails and the angle at which the nail enters the skin.

ABNORMAL PATTERN	EXACT DEFINITION & WAVEFORM FEATURE	COMMON CAUSES
1. Cheyne–Stokes Respiration	Cyclical abnormal pattern characterized by **gradually increasing depth and rate (hyperpnea), followed by a gradual decrease, and then a period of apnea**; cycle repeats.	**Neurological:** Stroke, brain injury, ↑ ICP. **Cardiac:** Congestive heart failure (esp. left-sided). **Other:** High altitudes, end-of-life (dying), CO poisoning.
2. Biot's Breathing	**Groups of quick, shallow inspirations followed by regular or irregular periods of apnea** (irregular in rhythm and depth, unlike smooth waxing/waning of Cheyne–Stokes).	Brainstem injury (esp. **medulla oblongata**), ↑ ICP, meningitis (esp. infants/children), head trauma, opioid overdose or CNS depression.
3. Kussmaul's Respiration	**Deep, labored, and rapid breathing pattern** — often described as **"air hunger"** — occurring as a **compensatory mechanism for metabolic acidosis**.	**Diabetic ketoacidosis (DKA)**, renal failure (uremic acidosis), severe lactic acidosis, toxic ingestions (salicylates, methanol, ethylene glycol).
4. Apneustic Breathing	**Prolonged inspiration (breathing in and holding the breath) followed by a short, inadequate expiration** ("gasping inspiration with a pause before expiration").	Brainstem injury (**pons** or upper medulla), stroke affecting the pons, severe head trauma, ↑ ICP, hypoxic brain injury.

2. IV. Renal System: Urine Output (Most Important Bedside Indicator) & Characteristics

URINE OUTPUT CLASSIFICATION & CALCULATION EXAMPLE

— **Normal urine output = 0.5 – 1 ml/kg/hour** (Adequate perfusion).

— **> 1 ml/kg/hr — Polyuria:** Diuretics, diabetes insipidus, fluid overload.

— **< 0.5 ml/kg/hr — Oliguria:** Poor perfusion – investigate.

— **< 0.3 ml/kg/hr — Severe oliguria:** Urgent medical review.

— **< 100 ml/day — Anuria:** Emergency – total failure or obstruction.

— **Calculation Example (Weight = 70 kg):** Minimum acceptable = **70 × 0.5 = 35 ml/hr** → Over 4 hrs = **140 ml minimum** (if passed 80 ml in 4 hrs → **Oliguria – report it**).

URINE CHARACTERISTICS & RENAL LABS

— **Colour:** Normal = **Pale yellow / straw** | Abnormal: **Dark** = dehydration; **Red** = blood; **Brown** = rhabdomyolysis; **Cloudy** = infection.

— **Clarity:** Normal = **Clear** | Abnormal: **Cloudy** = infection, protein, pus.

— **Odour:** Normal = **Mild** | Abnormal: **Offensive** = infection; **Sweet** = ketones.

— **Specific Gravity:** Normal = **1.005 – 1.030** | ↑ Dehydration; ↓ Overhydration / renal failure.

— Also assess: Fluid volume status, urinalysis (glucose, protein, pus, blood), genitalia, serum electrolytes, BUN, creatinine, and osmolarity.

تقييم الجهاز التنفسي والجهاز الكلوي (Respiratory & Renal Systems):

• **الجهاز التنفسي وأنماط التنفس غير الطبيعية:** الزرقة (Cyanosis) علامة متأخرة لنقص الأكسجين وأدق موضع لملاحظتها هو **اللسان والأغشية المخاطية**، بينما تعجر الأصابع (Clubbing) علامة لنقص الأكسجين المزمن. وتشمل أنماط التنفس غير الطبيعية: (1) **Cheyne–Stokes:** تزايد تدريجي في العمق والسرعة ثم تناقص تدريجي يعقبه توقف (Apnea) في السكتة الدماغية وفشل القلب الأيسر؛ (2) **Biot's:** مجموعات شهيق سريعة سطحية يتبعها توقف غير منتظم (إصابة النخاع المستطيل Medulla والتهاب السحايا)؛ (3) **Kussmaul's:** تنفس عميق وسريع ومجهد («جوع الهواء») كآلية تعويضية للحماض الاستقلابي مثل **DKA** والفشل الكلوي؛ (4) **Apneustic:** شهيق طويل مع حبس النفس يعقبه زفير قصير غير كاف (إصابة الجسر Pons).

• **الجهاز الكلوي وإدرار البول:** المعدل الطبيعي للبول **0.5–1 مل/كجم/ساعة**؛ أقل من **0.5 مل/كجم/ساعة = قلة بول (Oliguria)**، وأقل من **0.3 = قلة بول شديدة**، وأقل من **100 مل/يوم = انقطاع البول (Anuria)**. والكثافة النوعية الطبيعية **1.005–1.030** (ترتفع في الجفاف وتنخفض في فرط السوائل أو الفشل الكلوي).

6. V. Gastrointestinal & VI. Integumentary Systems (Braden, PUSH & SKIN)

1. V. Gastrointestinal (GI) Assessment & Stress Ulcers

ORDER OF ABDOMINAL EXAM & BOWEL SOUNDS

— **Mandatory Sequence: 1. Inspection → 2. Auscultation (Always BEFORE percussion and palpation) → 3. Percussion → 4. Palpation**.

— **How to Auscultate:** (1) Use **diaphragm** of stethoscope, (2) Listen in **all four quadrants**, (3) Listen for **at least 1 minute** before declaring absent, (4) Note character, frequency, and pitch.

— **Bowel Sound Classification:**
· **Normal (5–30/min):** Gurgling, clicking sounds (normal bowel activity).
· **Hyperactive:** Loud, frequent, high-pitched (diarrhea, early obstruction, hunger, GI bleed).
· **Hypoactive:** Reduced, infrequent (ileus, post-surgery, peritonitis).
· **Absent (no sounds after 5 minutes):** Paralytic ileus, peritonitis, late obstruction.
📌 *Note: the source says listen for at least **1 minute** before declaring absent, but the table defines absent as no sounds after **5 minutes**.*

STRESS ULCERS AND GI BLEEDING

— **Definition & Sites:** Mucosal erosions or ulcers that develop as a result of critical illness; common sites: **Stomach & Duodenum**.

— **Risk Factors: Mechanical ventilation, Coagulopathy, Burns, Head injury**.

— **Signs: Hematemesis, Melena, Blood in NGT aspirate, Falling hemoglobin (Hb)**.

— **Prevention:**
· **Proton Pump Inhibitors (PPIs)**
· **H_2-receptor blockers**
· **Early enteral feeding**

2. VI. Integumentary System: Braden Scale, PUSH Tool 3.0 & SKIN Bundle

Braden Risk Assessment Scale (6 Domains): 1. **Sensory** (1–4), 2. **Moisture** (1–4), 3. **Activity** (1–4), 4. **Mobility** (1–4), 5. **Nutrition** (1–4), 6. **Friction/Shear** (1–3).

BRADEN SCORE	RISK LEVEL	RECOMMENDED NURSING ACTION
19 – 23	**No Risk**	Routine skin care and reassessment
15 – 18	**Mild Risk**	Increase turning frequency – every 2 hours – moisture management – nutritional support
13 – 14	**Moderate Risk**	Pressure redistributing mattress – repositioning schedule – nutritional support – moisture management
10 – 12	**High Risk**	Specialty mattress – strict repositioning – protect heels – nutritional support – moisture management
≤ 9	**Very High Risk**	Highest level pressure redistribution surface – strict repositioning every 1–2 hours – heel protection – nutritional support – full skin care protocol

PRESSURE ULCER SCALE FOR HEALING (PUSH TOOL 3.0)

— Describes and follows pressure-ulcer healing using **3 main characteristics**:

— **1. Wound Size (Length × Width in cm²):** Scored **0 to 10** (0 = 0 cm^2 to 10 = > 24.0 cm^2).

— **2. Exudate Amount:** Scored **0 to 3** (0 None, 1 Light, 2 Moderate, 3 Heavy).

— **3. Tissue Type (Scored 0 to 4):**
· **0 – Closed:** Completely covered with epithelium (new skin).
· **1 – Epithelial Tissue:** Superficial ulcers, new pink or shiny tissue.
· **2 – Granulation Tissue:** Pink or beefy red tissue with shiny, moist granular appearance.
· **3 – Slough:** Yellow or white tissue adhering in strings or thick clumps.
· **4 – Necrotic Tissue (Eschar):** Black, brown, or tan tissue adhering firmly to wound bed.

THE SKIN BUNDLE (4 PREVENTION ELEMENTS)

— **S – Surface:** Correct mattress selection, use a **single layer of linen**, avoid wrinkles, do **not** place equipment or caps under the patient.

— **K – Keep Turning:** Reposition patient **every two hours (minimum)**, shift weight **every hour** when in a chair, use a pressure-relieving pad on the chair.

— **I – Incontinence Management:** Provide toileting assistance every 2 hours, perineal care every 2 hours & PRN, apply moisture barrier after perineal care, **avoid diapers**.

— **N – Nutrition:** Arrange nutrition consultation if at risk, carry out nutrition orders, encourage oral (PO) intake, record meals, supplements & fluid intake.

تقييم الجهاز الهضمي والجهاز الجلدي (GI & Integumentary Systems):
• **الجهاز الهضمي وقرح الإجهاد (Stress Ulcers):** الترتيب الإلزامي لفحص البطن هو: **Inspection → Auscultation → Percussion → → (دائمًا قبل القرع والجس) Palpation**. أصوات الأمعاء الطبيعية **5–30/دقيقة**، وتعتبر غائبة بعد الاستماع لمدة **5 دقائق**. وتحدث قرح الإجهاد في المعدة والاثني عشر بسبب التنفس الصناعي، اعتلال التجلط، الحروق، وإصابات الرأس، ويُوقى منها بمثبطات مضخة البروتون (**PPIs**) وحاصرات H_2 **والتغذية المعوية المبكرة**.
• **مقياس برادن (Braden Scale)، أداة PUSH، وحزمة SKIN:** يتكون مقياس برادن من **6 محاور** (19–23 لا يوجد خطر، 15–18 خطر بسيط، 13–14 متوسط، 10–12 مرتفع، ≥ **9 خطر مرتفع جدًا**). وتتابع أداة **PUSH** التئام القرحة عبر 3 خصائص: **حجم الجرح، كمية الإفرازات، ونوع الأنسجة (0 مغلق إلى 4 نسيج نخري أسود Eschar)**. وتشمل حزمة **SKIN**: السطح (**Surface**)، التقليب المستمر كل ساعتين بالسرير وكل ساعة بالكرسي (**Keep Turning**)، التحكم في السلس وتجنب الحفاضات (**Incontinence Management**)، والتغذية (**Nutrition**).

7. Ongoing Assessment, Care Plan, Clinical Case & Review MCQs

1. Fourth: Ongoing Assessment, Nursing Care Plan & Clinical Case Example

FOURTH: ONGOING ASSESSMENT & CARE PLAN

— **Ongoing Assessment:** Continuous monitoring during ICU stay to determine trends, evaluate response to therapy, and identify new problems. Frequency depends on stability: **every few minutes for extremely unstable patients to every 2 hours for very stable patients**. The nurse frequently rechecks **vital signs, ventilator & hemodynamic parameters, urine output, and laboratory trends**; changes (e.g., **new fever, drop in BP, change in mental status**) are recognized early through serial assessments; **psychosocial factors (anxiety, pain, family needs)** are re-evaluated regularly.

— **Nursing Care Plan Diagnoses & Interventions:**
· **Ineffective Airway Clearance:** Suctioning, repositioning, or adjusting ETT.
· **Impaired Gas Exchange (low SpO_2):** Increasing O_2 delivery, adjusting ventilator settings, monitoring ABGs.
· **Decreased Cardiac Output:** IV fluid bolus, titration of vasopressors, continuous hemodynamic monitoring.
· **Anxiety (Psychosocial):** Information, emotional support, possibly sedation.
· **Measurable Outcomes:** Maintain **SpO_2 > 92%**, **MAP ≥ 65 mmHg**.
· The care plan is **individualized**, **prioritized by severity**, and **regularly updated** as new assessment data emerge.
· **Care plan format:** Nursing diagnosis (unhealthy response related to contributing factor + subjective/objective data) → Nursing objective (short- & long-term goals) → Nursing implementation (assessment, interventions, health education) → Rationale → Evaluation of objective.

CLINICAL CASE EXAMPLE (68-YO MAN WITH PNEUMONIA & SEPTIC SHOCK)

— **Presentation:** Brought from ED with severe pneumonia and septic shock; hypotensive (**BP 70/40 mmHg**) and confused.

— **1. Pre-Arrival:** On broad-spectrum antibiotics & norepinephrine; nurse prepares ventilator, suction, IV fluids, vasopressors, and alerts ICU team.

— **2. Admission Quick Check (ABCDE):** Airway secured with ETT; Breathing labored, **SpO_2 84% on 100% FiO_2**, diffuse crackles; Circulation **BP 75/45, HR 130, cold skin**; Disability **GCS 6T**; Exposure **Temp 39.5°C** → increase ventilator/O_2, fluid bolus, titrate vasopressors, start antibiotics.

— **3. Comprehensive Assessment:** History of COPD & HTN; bilateral crackles; accessory muscle use; abdomen soft; mottled skin; pupils equal and reactive; sedation titrated for comfort; elevated lactate, leukocytosis, reduced urine output → **Nursing diagnoses:** ineffective breathing pattern, impaired gas exchange, decreased cardiac output, risk of infection.

— **4. Ongoing Assessment:** Frequent ABGs/ventilator checks, hourly hemodynamics & urine output, pain, agitation & sedation levels every shift & family updates/support.

2. High-Yield Self-Evaluation Review Questions

1 **During the Admission Quick Check (Primary Survey) of a critically ill patient with a non-patent airway, what flow rate of oxygen is administered after opening the airway?**

- ◯ a) 2 – 4 L/min
- ◯ b) 6 – 8 L/min
- ◯ c) 10 L/min
- ◯ **d) 15 L/min ✓**

الإجابة الصحيحة: d) 15 L/min — في جدول المسح الأولي ABCDE (ص 11)، عند التعامل مع مجرى الهواء غير السالك يتم فتح المجرى والتشفيط وإعطاء أكسجين بمعدل 15 L/min.

2 **Which abnormal pupil finding is classically associated with opioid overdose or a pontine lesion?**

- ◯ **a) Bilateral pinpoint pupils ✓**
- ◯ b) Unilateral fixed and dilated pupil
- ◯ c) Oval pupils
- ◯ d) Hippus phenomenon

الإجابة الصحيحة: a) Bilateral pinpoint pupils — الحدقات الدبوسية الثنائية الصغيرة جدًا والمتساوية تنتج عن جرعة زائدة من الأفيونات (Opioid overdose) أو إصابة الجسر (Pontine lesion).

3 **On the Behavioral Pain Scale (BPS), what score threshold identifies significant pain in a mechanically ventilated ICU patient?**

- ◯ a) BPS score ≥ 3
- ◯ b) BPS score ≥ 4
- ◯ **c) BPS score ≥ 6 ✓**
- ◯ d) BPS score ≥ 10

الإجابة الصحيحة: c) BPS score ≥ 6 — يتراوح مجموع مقياس BPS بين 3 و12، وتُعد الدرجة ≤ 6 دالة على وجود ألم ذي أهمية إكلينيكية (Significant pain).

4 **A 70-kg ICU patient has passed 80 mL of urine over the past 4 hours. How should the nurse interpret this finding?**

- ◯ a) Normal urine output (adequate perfusion)
- ◯ b) Polyuria
- ◯ **c) Oliguria (< 0.5 ml/kg/hr — minimum acceptable is 140 mL in 4 hours) ✓**
- ◯ d) Anuria

الإجابة الصحيحة: **c) Oliguria** — الحد الأدنى المقبول لمريض وزنه 70 كجم هو 70 × 0.5 = 35 مل/ساعة (أي 140 مل خلال 4 ساعات)، لذا فإن إخراج 80 مل في 4 ساعات يُعد قلة بول (**Oliguria**) تستوجب الإبلاغ الفوري.

5 **What is the correct sequence for physical examination of the abdomen in a critically ill patient?**

- ◯ a) Inspection → Palpation → Percussion → Auscultation
- ◯ **b) Inspection → Auscultation → Percussion → Palpation ✓**
- ◯ c) Auscultation → Inspection → Palpation → Percussion
- ◯ d) Palpation → Auscultation → Inspection → Percussion

الإجابة الصحيحة: **b) Inspection → Auscultation → Percussion → Palpation** — يجب دائمًا إجراء التسمع (**Auscultation**) قبل القرع والجس حتى لا تتغير حركية الأمعاء وأصواتها.

Closing

كفارة المجلس

سُبْحَانَكَ اللَّهُمَّ وَبِحَمْدِكَ، أَشْهَدُ أَنْ لَا إِلَهَ إِلَّا أَنْتَ، أَسْتَغْفِرُكَ وَأَتُوبُ إِلَيْكَ

دعاء بعد المذاكرة

اللَّهُمَّ إِنِّي أَسْتَوْدِعُكَ مَا عَلَّمْتَنِيهِ، فَارْدُدْهُ إِلَيَّ عِنْدَ حَاجَتِي إِلَيْهِ، وَلَا تَنْسَنِيهِ يَا رَبَّ الْعَالَمِينَ

ختامًا: نسأل الله أن يجعله **علمًا نافعًا**، وأن يوفقك في الامتحان ويكرمك بالنتيجة **التي ترضاها** يا رب العالمين. آمين